Implementation

1. Initialisation of Data Structures

Several data structures are initialised to represent and manage the locker system:

- **List (lockers)** – stores all locker records.
- **Set (used_phones)** – stores the unique mobile numbers associated with currently occupied lockers.
- **Dictionary (phone_usage, pending_discount, discount_codes)** –track customer usage, manage pending discounts, and store generated discount codes.
- **List (logs)** – stores activity records in chronological order.
- **Tuple (menu)** – stores fixed menu options.

2. Creation of Locker Data Structure

A loop (for i in range(1, 6)) creates five lockers. Each locker is represented by a dictionary containing the fields id, occupied, mobile, passcode, start_time, and paid_hours. The locker records are stored in a list, creating the central data structure used throughout the system. Each locker is represented by a dictionary because it allows information to be accessed using clear labels such as "mobile" and "passcode". This makes the program easier to understand and simplifies the process of updating and retrieving locker information.

3. Populating the Data

The lockers are populated with valid hardcoded values, including occupancy status, mobile numbers, passcodes, start times and paid_hours. Additional activity logs and customer tracking data are also created to support testing and to demonstrate functionality that will later be required by the locker simulator developed in Part 3.

4. Operations on Data Structures

The following operations demonstrate that the selected data structures support data storage, retrieval, updating, and automatic management of locker records:

- **Storage**: locker records are stored as dictionaries within the lockers list.
- **Retrieval (find_locker(locker_list, locker_id))**: find_locker() searches the list and returns a locker record for a specified locker ID.
- **Updating (update_tracking())**: updates the used_phones set and phone_usage dictionary using data stored in the locker records.

- **Display (print_state())**: locker information can be displayed individually or collectively using loops.
- **Automatic management (free_expired(current_time))**: free_expired() updates locker records when their rental period has expired.

Additional supporting structures are used to manage customer tracking, discount information, activity logs, and menu options.

Justification of Data Structures and Data Types

A list was selected as the primary collection because locker records must be stored in sequence and accessed repeatedly when searching, updating and displaying locker information.

Each locker stores the following fields:

- *id (string):* Used for locker identifiers such as "L01". Strings support alphanumeric values and consistent formatting.
- *occupied (boolean):* Represents whether a locker is in use as True or False.
- *mobile (string):* Mobile numbers are stored as strings because they are not used for arithmetic calculations and may contain leading zeros.
- *passcode (string):* Strings allow numeric or alphanumeric passcodes without restriction.
- *start_time (integer):* Stored as an integer to support expiry calculations.
- *duration (integer):* Represents how long a locker is occupied and allows efficient time calculations.
- *Set* (used_phones): suitable for enforcing uniqueness because duplicate values are automatically rejected.
- *Dictionaries* (phone_usage, pending_discount, discount_codes) store phone numbers as keys and usage counts as values, enabling fast lookup and updates.
- *List* (logs) stores activity records in chronological order.

The selected data structures support efficient storage, retrieval, updating, and tracking of locker information, making them suitable for the requirements of the locker system.

Assumptions and Justifications:

- Five lockers are used to simplify system management and testing while still providing sufficient functionality to demonstrate all required scenarios for 10% and 20% discount.

- The system assumes only one user interacts with it at a time, in a single console environment without multiuser support.
- The system allows a single customer to rent multiple lockers using the same mobile number. Linking multiple lockers to a single customer provides greater flexibility for users who need additional storage space. At the same time, the system can easily identify and manage all lockers associated with the same mobile number.
- The mobile number is used as a user identifier, because it is unique, easy for users to remember, and can also be used for receipt delivery, verification, and activity tracking. Storing mobile numbers as strings can preserve formatting and avoid issues associated with numeric processing of identifiers.
- A 4-digit passcode is used to ensure a simple but effective level of security while maintaining ease of use for customers. Authentication mechanisms based on a passcode improve protection against unauthorised access and are commonly used in locker and access-control systems.
- Fixed hourly rates are used for pricing. This simplifies pricing calculations and ensures predictable costs for users. The tiered charging structure also enables the system to demonstrate different charging and penalty scenarios.
- A penalty is applied if the locker is used beyond maximum allowed rented duration. This reflects real-world locker policies, compensates for extended occupancy, and encourages users to collect their belongings on time.
- All payments are assumed to be successful once processed. The system simulates payment and does not connect to a real payment gateway.
- SMS verification codes are simulated using a fixed verification code. This simplifies implementation and avoids reliance on external communication systems while demonstrating passcode recovery functionality.
- The Smart Locker System is assumed to operate as part of a wider locker network. As a result, discount codes earned at one location can be redeemed at another location within the network.
- Data is not stored permanently. The program does not use files or databases, so all data resets when it ends.

Design choices and Justifications

The program is structured into three main parts: global data structures, functions, and a main while loop. First part stores locker, customer, and system information. The functions perform reusable tasks such as validation, charge calculation, payment processing, authentication, and activity logging.

While the main while loop controls the menu-driven interaction and allows the system to continue running until the user chooses to exit. Figure 14 illustrates the overall program flow.

The following points justify the design choices made in the system:

- Time is simulated using an internal counter that advances after each completed menu cycle. This approach was chosen instead of asking the user to manually enter elapsed hours, as it provides a more realistic simulation of a locker system and better user experience where time progresses automatically. It also enables storage charges, penalties, and confiscation rules to be demonstrated and tested efficiently.
- The payment process is divided into two stages. An initial payment is collected during the storage stage to secure the locker reservation and guarantee payment for the basic storage service. A second payment is only needed if the user exceeds the allocated storage period. This approach reduces the risk of unpaid locker usage and improves locker availability.
- Receipts are generated after each payment transaction to provide proof of payment and allow users to verify storage and overstay charges.
- Simulated timestamps are used for receipts. This enables realistic receipt output without relying on external date and time modules.
- A main *while loop* is used to control program flow. This allows the system to run continuously and repeatedly display the menu until the user exits.
- The menu is stored as a tuple. This provides a fixed and structured representation of the user interface.
- A *match-case* statement is used to handle menu selections. Each menu option is implemented in a separate case, providing a clear and structured alternative to multiple if-elif statements.
- A *list of dictionaries* is used to store locker data. This structure allows multiple attributes to be organised and accessed easily.
- *Dictionaries* are used to store user-related data such as phone usage and discounts. They were preferred over lists because values can be accessed directly using a mobile number as a key, avoiding the need to search through multiple records. This enables faster retrieval and updating of customer information
- Sets are used to store phone numbers, ensuring uniqueness and preventing duplicate entries.
- Functions are used to promote modularity. Functions improve code reuse, readability, and maintainability in software development.
- Input validation functions are implemented for all user inputs. This ensures data accuracy and prevents invalid input from causing errors. Input validation is an important programming practice because it improves reliability and reduces the likelihood of errors caused by invalid user input.

- Exception handling is implemented in payment processing and duration validation. This prevents program crashes when users enter invalid values.

Implementation

The system provides a menu-driven interface with options to view lockers, store luggage, retrieve luggage, my usage info, and exit. When storing luggage, the system first checks if lockers are available. If all lockers are occupied, the user is notified immediately. A high-demand warning is displayed as well in Option 1 when three or more lockers are occupied (Figures 15 and 16).

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 1

=== LOCKER STATUS ===
L01   |  Occupied
L02   |  Occupied
L03   |  Occupied
L04   |  Occupied
L05   |  Occupied
⚠All lockers are occupied!

Currently active users: 1

=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 2
All lockers are currently occupied.
```

Figure 1. System Response When All Lockers Are Occupied (Options 1 and 2)

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 1

=== LOCKER STATUS ===
L01   |  Occupied
L02   |  Occupied
L03   |  Occupied
L04   |  Available
L05   |  Available
⚠High demand: Most lockers are occupied!
```

Figure 2. High Demand Alert When Most Lockers Are Occupied

If a locker is available, the user selects a locker ID and enters a validated mobile number, passcode, and rental duration. The system then calculates and displays the storage charge before asking the user to confirm the booking. After confirmation, the customer is given the opportunity to enter a discount code obtained from another location (Figure 17 and 18).

```
Enter number of hours: 6
Estimated cost: £17.50
Rate 1 and Rate 2 charges applied.
Note: After 15 hours penalty rate will be applied.
Late collection (after 24 hours) may result in
 additional charges or luggage confiscation.

Confirm storing luggage? (Y/N): y
Enter discount code (if you have one, otherwise press Enter): DISC2-789
Discount code accepted!
Final price to pay: £15.75
```

Figure 3. Redeeming a 10% Discount Using a Discount Code

```
Enter locker ID: l05
Enter mobile number: 07123456789
Enter 4-digit passcode: 1234

Enter number of hours: 5
Estimated cost: £12.50
Rate 1 charge applied.
Note: After 5 hours higher rate will apply.

Confirm storing luggage? (Y/N): y
Enter discount code (if you have one, otherwise press Enter): DISC4-789
Discount code accepted!
Final price to pay: £10.00
```

Figure 4. Redeeming a 20% Discount Using a Discount Code

If no valid discount code is entered, the system automatically checks the customer's usage history and applies any eligible loyalty rewards (Figure 19 and 20). Customers who have used the locker system multiple times may receive automatic discounts. Customers who become eligible for a 10% discount may choose to apply it immediately or save the reward to unlock a larger 20% discount after additional usage (Figure 21).

```
Enter number of hours: 6
Estimated cost: £17.50
Rate 1 and Rate 2 charges applied.
Note: After 15 hours penalty rate will be applied.
Late collection (after 24 hours) may result in
 additional charges or luggage confiscation.

Confirm storing luggage? (Y/N): y
Enter discount code (if you have one, otherwise press Enter):

Loyalty reward: 10% discount available!
Discounted price: £15.75
Use 10% now (Y) or save for 20% later (N)? y
10% discount applied!
Final price to pay: £15.75
```

Figure 5. Automatic Application of a 10% Discount

```
Enter number of hours: 6
Estimated cost: £17.50
Rate 1 and Rate 2 charges applied.
Note: After 15 hours penalty rate will be applied.
Late collection (after 24 hours) may result in
 additional charges or luggage confiscation.

Confirm storing luggage? (Y/N): y
Enter discount code (if you have one, otherwise press Enter):

Loyalty reward: 20% discount applied!
Final price to pay: £14.00
```

Figure 6. Automatic Application of a 20% Discount

```
Enter number of hours: 6
Estimated cost: £17.50
Rate 1 and Rate 2 charges applied.
Note: After 15 hours penalty rate will be applied.
Late collection (after 24 hours) may result in
 additional charges or luggage confiscation.

Confirm storing luggage? (Y/N): y
Enter discount code (if you have one, otherwise press Enter):

Loyalty reward: 10% discount available!
Discounted price: £15.75
Use 10% now (Y) or save for 20% later (N)? n
Discount saved.

Keep going! 20% discount unlocks at 5 uses.
You can redeem your discount at any time using Option 4.
Final price to pay: £17.50
```

Figure 7. Saving a 10% Discount for a Future 20% Discount

Encouraging message displays when a customer chooses to save 10% discount for future use, to unlock a larger 20% discount reward (Figure 22).

```
Enter number of hours: 6
Estimated cost: £17.50
Rate 1 and Rate 2 charges applied.
Note: After 15 hours penalty rate will be applied.
Late collection (after 24 hours) may result in
 additional charges or luggage confiscation.

Confirm storing luggage? (Y/N): y
Enter discount code (if you have one, otherwise press Enter):

Saved discount pending. Keep going to reach 20%!
Final price to pay: £17.50
```

Figure 8. Saved Discount Pending Before the 5th Usage

After any applicable discount has been applied, the system displays the final amount due along with the available payment options. Once payment has been successfully completed, a receipt is generated showing details such as the locker ID, mobile number, payment amount, status, and transaction time (Figure 25). The customer is also given the option to receive the receipt via SMS (Figures 23). Figures 24–25 illustrate the normal storing process with payment procedure, and receipt generation.

```
Would you like a digital receipt sent to your mobile? (Y/N): y

Processing request...
Sending SMS...
Receipt sent to ********789

Locker L05 is now occupied.
Please keep your passcode safe.

Thank you for using the SMART LOCKER SYSTEM.
```

Figure 9. Digital Receipt Sent to the Customer's Mobile Number

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 1

=== LOCKER STATUS ===
L01   |  Available
L02   |  Available
L03   |  Available
L04   |  Available
L05   |  Available

Currently active users: 0

=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 2

Available lockers: L01, L02, L03, L04, L05

Enter locker ID: l01
Enter mobile number: 07123456789
Enter 4-digit passcode: 1234
Enter number of hours: 2
Estimated cost: £5.00
Rate 1 charge applied.
Note: After 5 hours higher rate will apply.

Confirm storing luggage? (Y/N): y
Enter discount code (if you have one, otherwise press Enter):

No discount applied.
Final price to pay: £5.00

Please pay to proceed.
1. Cash
2. Card
```

Figure 10. Normal Storage Process – Input and Cost Calculation

```
Enter option: 2

Insert card...
Processing...
Card accepted.
Payment of £5.00 received.

Door opened. Please store your luggage.

=========== RECEIPT ===========
Locker:     L01
Mobile:     07123456789
Status:     Stored
--------------------------------
Paid:       £5.00
Date:       2026-05-31
Time:       11:01:12
================================
Thank you for using our service!

Would you like a digital receipt sent to your mobile? (Y/N): n

No SMS will be sent.

Locker L01 is now occupied.
Please keep your passcode safe.

Thank you for using the SMART LOCKER SYSTEM.
Note:
-Maximum storage time: 15 hours
-Fee:
Rate 1: £2.5/hour (first 5 hours)
Rate 2: £5/hour (next 10 hours)
Rate 3: £10/hour (after 15 hours - penalty rate)
-Late collection: Items not collected after 24 hours may incur
 in additional charges or be confiscated.

=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 1

=== LOCKER STATUS ===
L01   |  Occupied
L02   |  Available
L03   |  Available
L04   |  Available
L05   |  Available

Currently active users: 1
```

Figure 11. Normal Storage – Card Payment and Confirmation

The system allows to cancel the storage process before confirming the booking (Figure 26).

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 2

Available lockers: L01, L02, L03, L04, L05

Enter locker ID: l02
Enter mobile number: 07123456789
Enter 4-digit passcode: 1234

Enter number of hours: 2
Estimated cost: £5.00
Rate 1 charge applied.
Note: After 5 hours higher rate will apply.

Confirm storing luggage? (Y/N): n
Operation cancelled.
```

Figure 12. Customer Cancelling the Storage Process

The retrieval stage starts with users' mobile number validation. Then the system searches for any active lockers linked to entered mobile number. If the mobile number rents multiple lockers, the user is prompted to select the locker they wish to open. The system then validates the passcode, calculates the storage duration, and displays any outstanding charges if existed before opening the locker. Figure 27 shows a successful retrieval with no additional charges, while Figure 28 illustrates the outcome when no active locker is found for the entered mobile number.

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 3
Enter mobile number: 07123456789

This mobile has rented multiple lockers:
- L01
- L02

Enter locker ID: l01
Enter 4-digit passcode: 1234

Time used: 3h
Time paid: 4h
No extra charge. Thank you!

No payment required. Unlocking...

Locker L01 opened successfully.
Thank you for using the SMART LOCKER SYSTEM.
```

Figure 13. Normal Retrieval – No Extra charge

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 3
Enter mobile number: 07987456123
No lockers found for this mobile number.
```

Figure 14. Retrieval Process Unsuccessful Due to an Incorrect Mobile Number

To improve security, the system restricts users to three passcode attempts during the retrieval process. Access is denied after three failed attempts, reducing the risk of unauthorised access to stored luggage (Figure 29).

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 3
Enter mobile number: 07123456789

This mobile has rented multiple lockers:
- L02
- L03
- L04
- L05

Enter locker ID to retrieve: l02
Enter 4-digit passcode: 2389
Incorrect passcode. You have 2 attempt(s) remaining.
Forgot passcode? (Y/N): n
Enter 4-digit passcode: 6547
Incorrect passcode. You have 1 attempt(s) remaining.
Forgot passcode? (Y/N): n
Enter 4-digit passcode: 9512
Incorrect passcode. You have 0 attempt(s) remaining.
Incorrect passcode. Too many attempts. Access denied. Try later.
```

Figure 15. Access Denied After Three Incorrect Passcode Attempts

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 3
Enter mobile number: 07123456789

This mobile has rented multiple lockers:
- L01
- L02
- L03
Enter locker ID to retrieve: l01
Enter 4-digit passcode: 5678
Incorrect passcode. You have 2 attempt(s) remaining.
Forgot passcode? (Y/N): n
Enter 4-digit passcode: 12345
Invalid passcode. Must be 4 digits.

Enter 4-digit passcode: 8912
Incorrect passcode. You have 1 attempt(s) remaining.
Forgot passcode? (Y/N): n
Enter 4-digit passcode: 1234

Time used: 7h
Time paid: 24h
No extra charge. Thank you!

No payment required. Unlocking...

Locker L01 opened successfully.
Thank you for using the SMART LOCKER SYSTEM.
```

Figure 16. Authentication Successful After Two Incorrect Passcode Attempts

The system includes a passcode recovery option for users who forget their passcode by sending a verification code to the registered mobile number. Figure 30 demonstrates the passcode recovery process.

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 3
Enter mobile number: 07123456789

This mobile has rented multiple lockers:
- L01
- L02
- L03
- L04
- L05

Enter locker ID to retrieve: l01
Enter 4-digit passcode: 2345
Incorrect passcode. You have 2 attempt(s) remaining.
Forgot passcode? (Y/N): y
A verification code has been sent to ********789.
Enter verification code: 12345
Enter new 4-digit passcode: 1234
Passcode reset successful.
Please log in again using your new passcode.
Enter 4-digit passcode: 1234

Time used: 9h
Time paid: 2h
⚠ Extra time: 7h
Extra charge: £35.00

Please pay £35 to unlock the locker.
```

Figure 17. Passcode Recovery Using a Verification Code After an Incorrect Attempt

The system uses a tiered charging structure based on storage duration. Customers select the required number of hours and make payment by card or cash, as demonstrated in Figures 32–36.

```
Confirm storing luggage? (Y/N): y
Enter discount code (if you have one, otherwise press Enter):

No discount applied.
Final price to pay: £5.00

Please pay to proceed.
1. Cash
2. Card

Enter option: 2
Processing...
Card accepted.
Payment of £5.00 received.

Door opened. Please store your luggage.

========== RECEIPT ==========
Locker:     L05
Mobile:     07123456789
Status:     Stored
------------------------------
Paid:       £5.00
Date:       2026-05-31
Time:       11:25:05
==============================
Thank you for using our service!
```

Figure 18. Card Payment

```
Confirm storing luggage? (Y/N): y
Enter discount code (if you have one, otherwise press Enter):

No discount applied.
Final price to pay: £47.50

Please pay to proceed.
1. Cash
2. Card

Enter option: 1
Enter cash amount inserted: £50
Insert cash...
Processing...
Cash accepted.
Please take your change: £2.50
Payment of £47.50 received.

Door opened. Please store your luggage.

=========== RECEIPT ===========
Locker:     L05
Mobile:     07123456789
Status:     Stored
--------------------------------
Paid:       £47.50
Date:       2026-05-31
Time:       11:25:05
================================
Thank you for using our service!
```

Figure 19. Cash Payment with Change

```
Confirm storing luggage? (Y/N): y
Enter discount code (if you have one, otherwise press Enter):

No discount applied.
Final price to pay: £47.50

Please pay to proceed.
1. Cash
2. Card
Enter option: 1
Enter cash amount inserted: £40
Insufficient cash. You still owe £7.50.
Press C to continue with cash or S to switch to card: s

You still owe £7.50.

Insert card...
Processing...
Card accepted.
Remaining amount £7.50 paid.
Total amount paid: £47.50

Door opened. Please store your luggage.

========== RECEIPT ==========
Locker:     L01
Mobile:     07123456789
Status:     Stored
------------------------------
Paid:       £47.50
Date:       2026-05-31
Time:       11:01:12
==============================
Thank you for using our service!
```

Figure 20. Cash Payment Combined with Card Payment

```
Confirm storing luggage? (Y/N): y
Enter discount code (if you have one, otherwise press Enter):

No discount applied.
Final price to pay: £47.50

Please pay to proceed.
1. Cash
2. Card
Enter option: 1
Enter cash amount inserted: £40
Insufficient cash. You still owe £7.50.
Press C to continue with cash or S to switch to card: c
Enter cash amount inserted: £10

Insert cash...
Processing...
Cash accepted.
Please take your change: £2.50
Payment of £47.50 received.

Door opened. Please store your luggage.

========== RECEIPT ==========
Locker:     L01
Mobile:     07123456789
Status:     Stored
------------------------------
Paid:       £47.50
Date:       2026-05-31
Time:       11:01:12
==============================
Thank you for using our service!
```

Figure 21. Two-Step Cash Payment with Change

```
Confirm storing luggage? (Y/N): y
Enter discount code (if you have one, otherwise press Enter):

No discount applied.
Final price to pay: £47.50

Please pay to proceed.
1. Cash
2. Card
Enter option: 1
Enter cash amount inserted: £40
Insufficient cash. You still owe £7.50.
Press C to continue with cash or S to switch to card: c
Enter cash amount inserted: £7.5

Insert cash...
Processing...
Cash accepted.
Exact amount received. No change.
Payment of £47.50 received.

Door opened. Please store your luggage.

========== RECEIPT ==========
Locker:     L01
Mobile:     07123456789
Status:     Stored
--------------------------------
Paid:       £47.50
Date:       2026-05-31
Time:       11:01:12
================================
Thank you for using our service!
```

Figure 22. Two-Step Cash Payment with No Change

During retrieval, the system compares the actual storage duration with the paid duration from storage stage and calculates any additional charges automatically. Any outstanding amount must be paid before the locker can be opened. Figures 37–41 demonstrate the calculation of additional charges and penalty rates for overdue storage periods. In each figure, the receipt on the left shows the original storage transaction, while the receipt on the right shows the retrieval transaction, including any additional charges or penalties applied, as well as the updated status and date of the transaction.

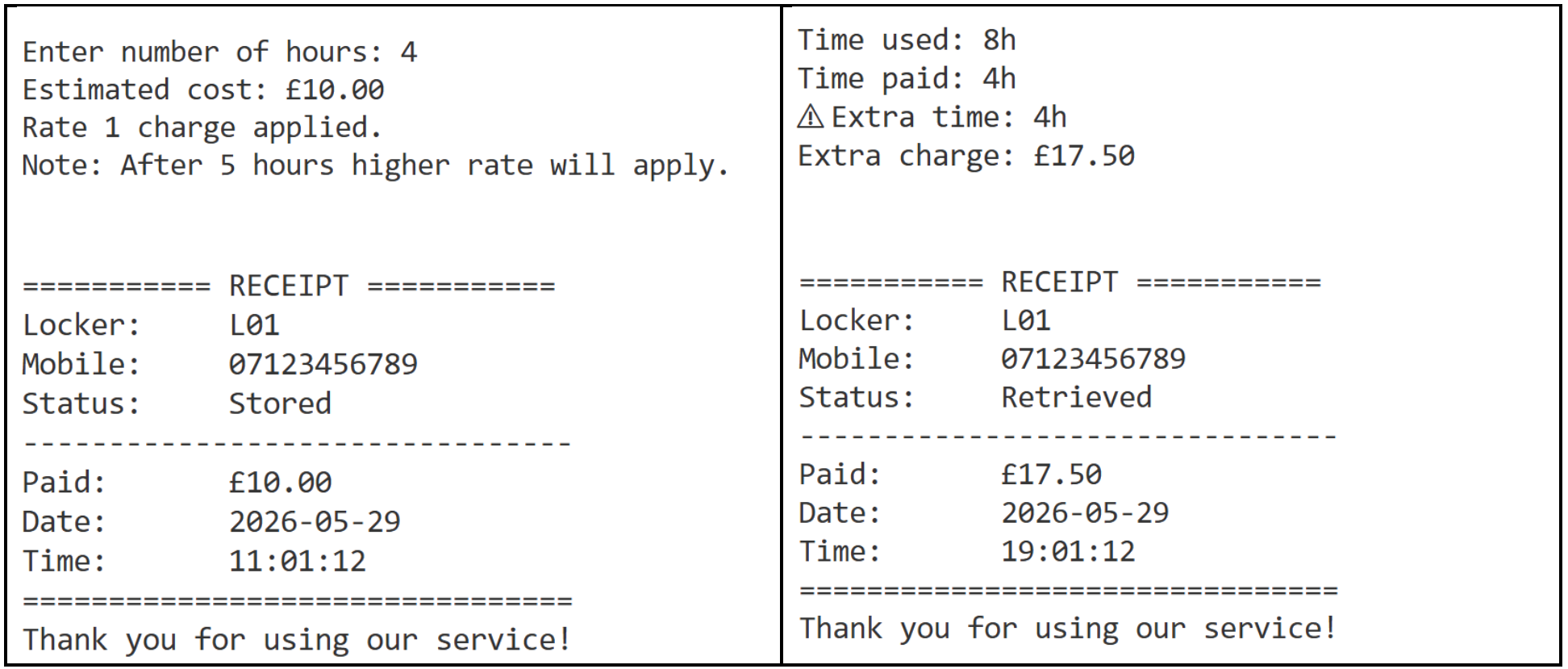


Figure 23. Rate 1 and Rate 2 Charges Applied for Additional Hours

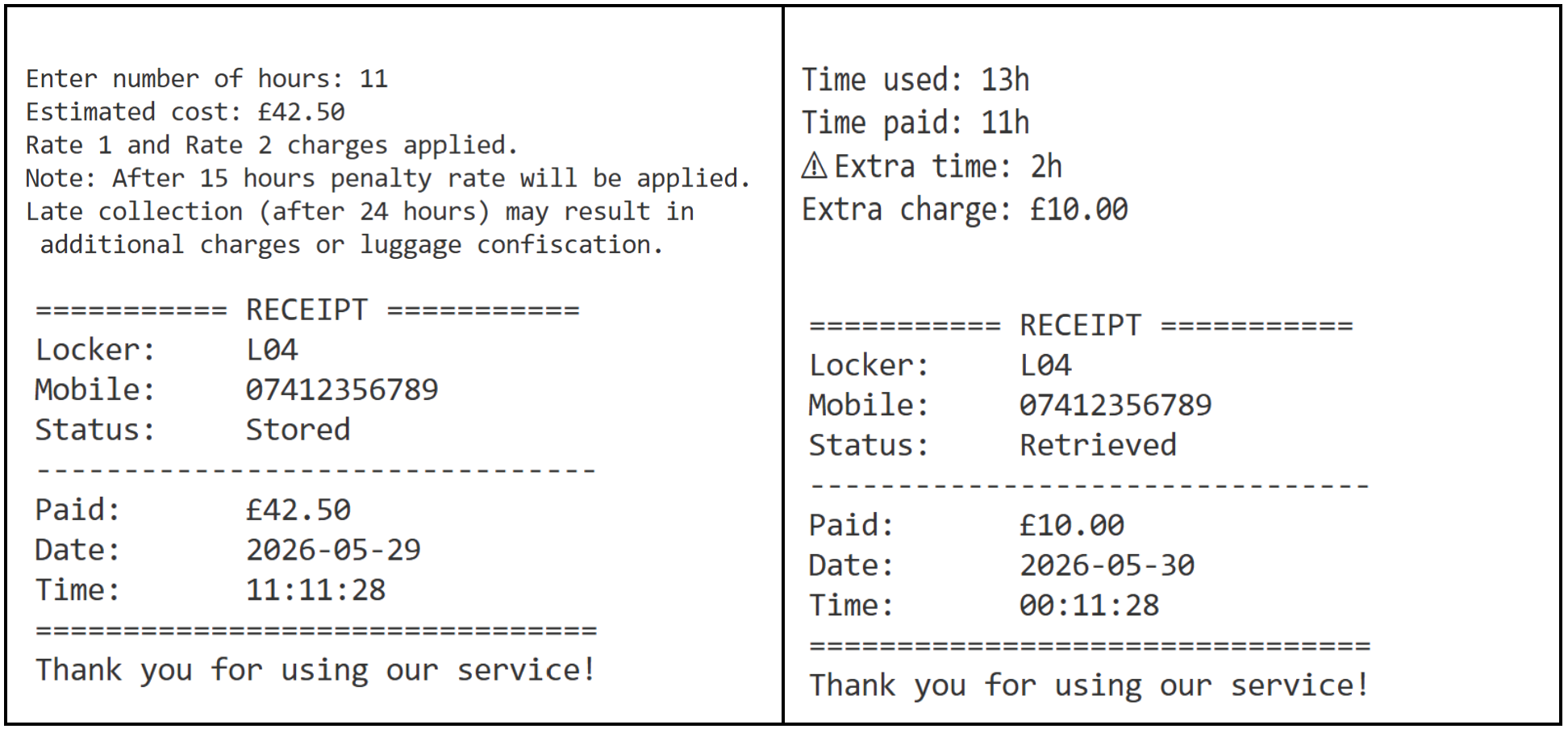


Figure 24. Rate 2 Charge Applied for Additional Hours

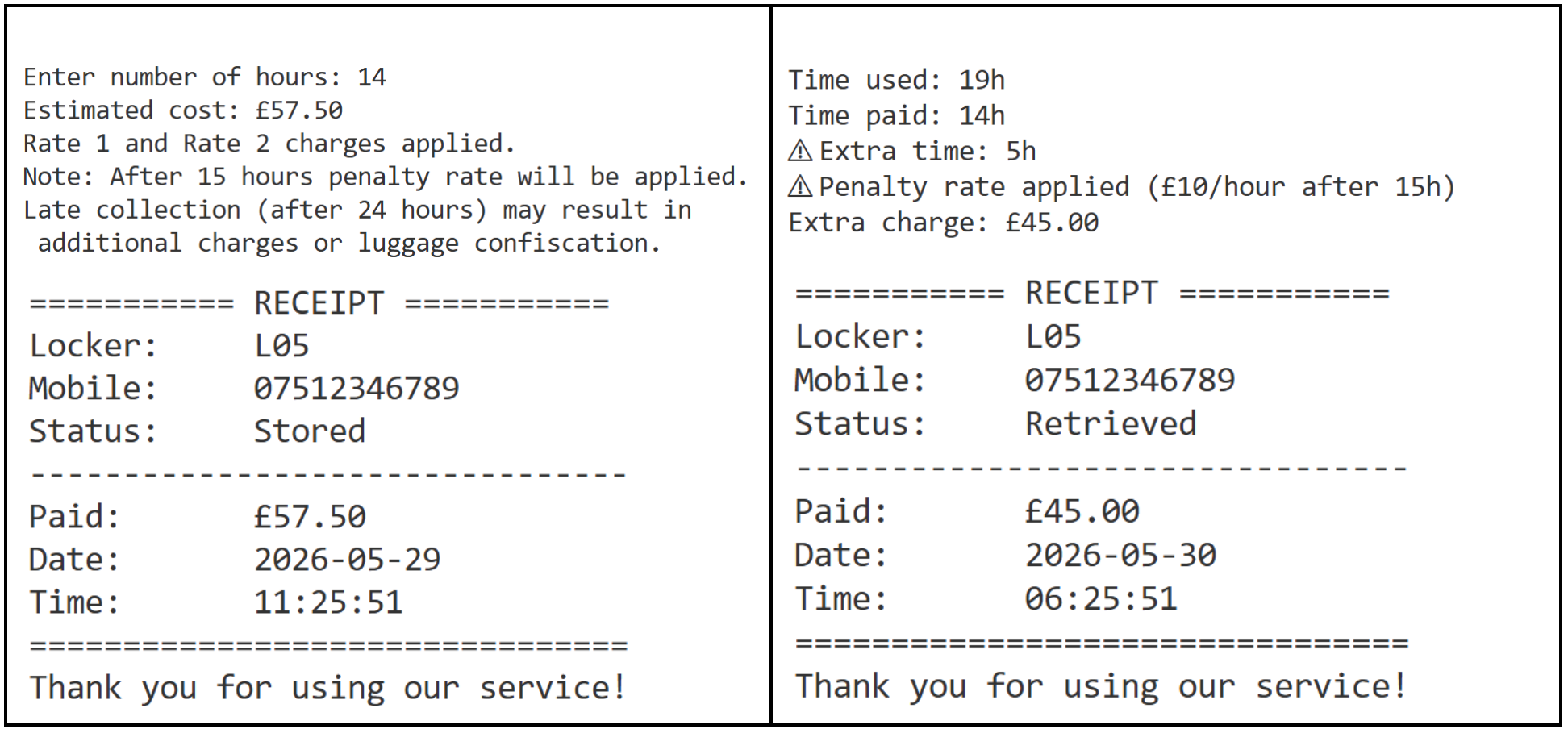


Figure 25. Rate 2 and Rate 3 (Penalty) Charges for Extra Storage Time

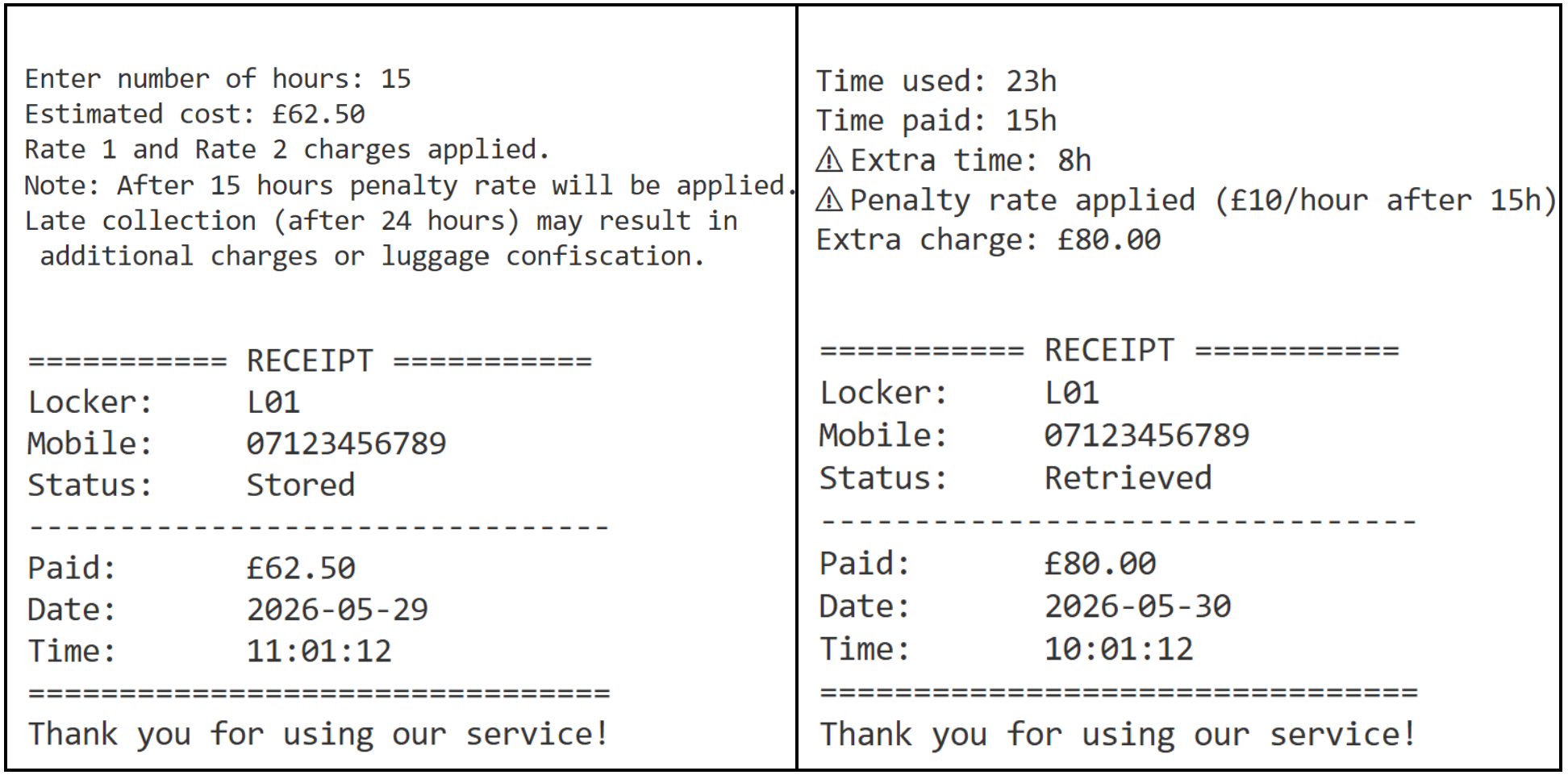


Figure 26. Rate 3 (Penalty) Applied for an Additional 8 Hours

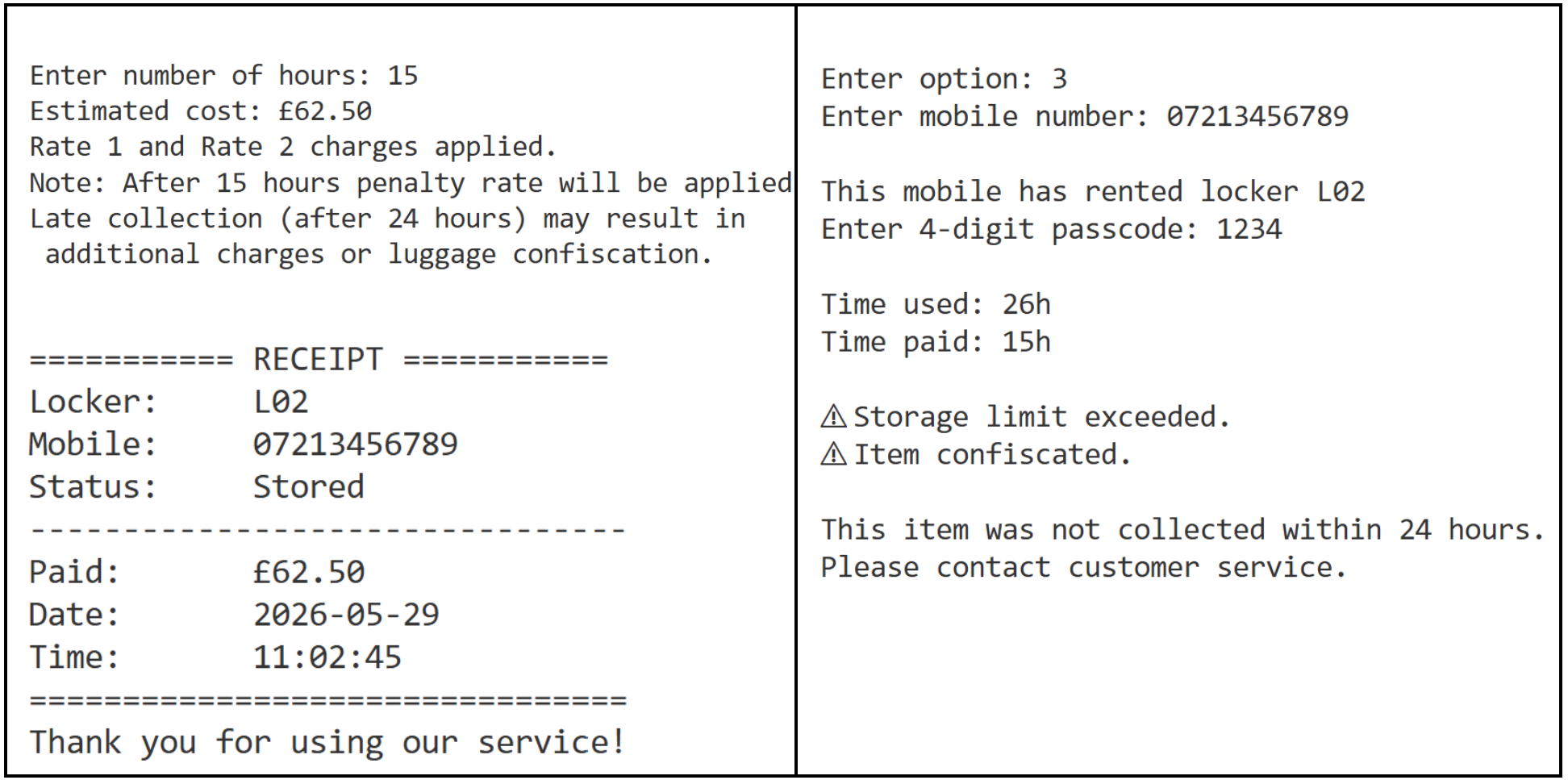


Figure 27. Confiscation of Uncollected Luggage

Various input validation checks have been implemented across the system to prevent incorrect data from being processed. These checks cover menu options, mobile numbers, locker IDs, passcodes, storage durations, payment methods, SMS requests, and passcode recovery inputs. When an invalid value is detected, the program requests a new input rather than continuing with incorrect data. Figures 42–51 provide examples of these validation checks during testing.

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 7
Invalid option. Please enter 1-5.

Enter option: 12
Invalid option. Please enter 1-5.

Enter option: q
Invalid option. Please enter 1-5.

Enter option: K
Invalid option. Please enter 1-5.

Enter option:   1

=== LOCKER STATUS ===
L01   |  Available
L02   |  Available
L03   |  Available
L04   |  Available
L05   |  Available

Currently active users: 0
```

Figure 28. Input Validation for Menu Options

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 3
Enter mobile number: 071234567899
Invalid mobile number. Must be 11 digits.
Enter mobile number: 0712345678w
Invalid mobile number. Must be 11 digits.
Enter mobile number:        07123456789
```

Figure 29. Input Validation for Mobile Number

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 2

Available lockers: L01, L02, L03, L04, L05

Enter locker ID: 123
Invalid locker ID. Choose from: L01, L02, L03, L04, L05

Enter locker ID: x01
Invalid locker ID. Choose from: L01, L02, L03, L04, L05

Enter locker ID: asd
Invalid locker ID. Choose from: L01, L02, L03, L04, L05

Enter locker ID: l01
Enter mobile number:
```

Figure 30. Input Validation for Locker ID

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 2

Available lockers: L02, L03, L04, L05

Enter locker ID: l02
Enter mobile number: 07123456789
Enter 4-digit passcode: 12345
Invalid passcode. Must be 4 digits.
Enter 4-digit passcode: m123
Invalid passcode. Must be 4 digits.
Enter 4-digit passcode: 123m
Invalid passcode. Must be 4 digits.
Enter 4-digit passcode: mnbv
Invalid passcode. Must be 4 digits.
Enter 4-digit passcode: 1234

Enter number of hours:
```

Figure 31. Input Validation for Passcode

```
Confirm storing luggage? (Y/N): d
Invalid input! Please enter Y or N only.

Confirm storing luggage? (Y/N): 12
Invalid input! Please enter Y or N only.

Confirm storing luggage? (Y/N):    y
Enter discount code (if you have one, otherwise press Enter):

No discount applied.
Final price to pay: £47.50
```

Figure 32. Input Validation for Storage Confirmation

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 2

Available lockers: L01, L02, L03, L04, L05

Enter locker ID: l01
Enter mobile number: 07123456789
Enter 4-digit passcode: 1234

Enter number of hours: 16
Invalid duration. Maximum storage time is 15 hours.

Enter number of hours:    15
Estimated cost: £62.50
Rate 1 and Rate 2 charges applied.
Note: After 15 hours penalty rate will be applied.
Late collection (after 24 hours) may result in
 additional charges or luggage confiscation.
```

Figure 33. Input Validation for Duration Entry in Hours

```
No discount applied.
Final price to pay: £47.50

Please pay to proceed.
1. Cash
2. Card

Enter option: 3
Invalid option. Enter 1 for Cash or 2 for Card.

Enter option: k
Invalid option. Enter 1 for Cash or 2 for Card.

Enter option:     2

Insert card...
Processing...
Card accepted.
Payment of £47.50 received.

Door opened. Please store your luggage.
```

Figure 34. Input Validation for Payment Method Options

```
Please pay to proceed.
1. Cash
2. Card

Enter option: 1
Enter cash amount inserted: £40
Insufficient cash. You still owe £7.50.
Press C to continue with cash or S to switch to card: 1
Invalid option. Please enter C or S.
Press C to continue with cash or S to switch to card: k
Invalid option. Please enter C or S.
Press C to continue with cash or S to switch to card: 12
Invalid option. Please enter C or S.
Press C to continue with cash or S to switch to card: jk
Invalid option. Please enter C or S.
Press C to continue with cash or S to switch to card:   s

You still owe £7.50.
```

Figure 35. Input Validation for Additional Payment Method Selection

```
Would you like a digital receipt sent to your mobile? (Y/N): 1
Invalid input. Please enter Y or N.

Would you like a digital receipt sent to your mobile? (Y/N): 12
Invalid input. Please enter Y or N.

Would you like a digital receipt sent to your mobile? (Y/N): k
Invalid input. Please enter Y or N.

Would you like a digital receipt sent to your mobile? (Y/N): kl
Invalid input. Please enter Y or N.

Would you like a digital receipt sent to your mobile? (Y/N):    n

No SMS will be sent.

Locker L01 is now occupied.
Please keep your passcode safe.

Thank you for using the SMART LOCKER SYSTEM.
```

Figure 36. Input Validation for Sending a Digital Receipt to a Mobile Number

```
Enter option: 3
Enter mobile number: 07123456789

This mobile has rented locker L01
Enter 4-digit passcode: 2345
Incorrect passcode. You have 2 attempt(s) remaining.
Forgot passcode? (Y/N): 2
Invalid input. Please enter Y or N.
Forgot passcode? (Y/N): w
Invalid input. Please enter Y or N.
Forgot passcode? (Y/N):     y
A verification code has been sent to ********789.
Enter verification code:
```

Figure 37. Input Validation for Forgotten Passcode Recovery

Option 4 allows customers to view their locker system usage history. Once a validated mobile number has been entered, the user can view their account information, including usage statistics, activity logs, history of storage, and retrieval. Information about loyalty rewards, available discounts, and total usage is also provided (Figures 52–54).

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 4
Enter mobile number: 07412356789

Total uses: 0

=== YOUR ACTIVITY LOG ===
No activity found

No discount available right now.
3 more use(s) to next 10% reward.
5 more use(s) to next 20% reward.
```

Figure 38. No Activity Log Found for an Incorrect Mobile Number

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 4
Enter mobile number: 07123456789

Total uses: 5

=== YOUR ACTIVITY LOG ===
L01 | stored by 07123456789
L02 | stored by 07123456789
L03 | stored by 07123456789
L04 | stored by 07123456789
L05 | stored by 07123456789
L01 | retrieved by 07123456789

No discount available right now.
3 more use(s) to next 10% reward.
5 more use(s) to next 20% reward.
```

Figure 39. Activity Log for a Frequent Customer

```
Enter option: 4
Enter mobile number: 07123456789

Total uses: 2

=== YOUR ACTIVITY LOG ===
L01 | stored by 07123456789
L02 | stored by 07123456789

Next booking will have a 10% discount.
Your discount code: DISC2-789
You can use it at this terminal automatically or redeem it at another station.
```

Figure 40. Generating a 10% Discount Code for Use Across Multiple Locations

```
Enter option: 4
Enter mobile number: 07123456789

Total uses: 4

=== YOUR ACTIVITY LOG ===
L01 | stored by 07123456789
L02 | stored by 07123456789
L03 | stored by 07123456789
L04 | stored by 07123456789

Almost there!
Your NEXT booking will have a 20% discount.
Your discount code: DISC4-789
You can use it at this terminal automatically or redeem it at another station.
```

Figure 41. Generating a 20% Discount Code for Use Across Multiple Locations

Figures 56-58 demonstrate the use of exception handling to manage invalid user input and prevent program termination.

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 2

Available lockers: L03, L04, L05

Enter locker ID: l03
Enter mobile number: 07123456789
Enter 4-digit passcode: 1234

Enter number of hours: ^Z
Handled error: No input provided.
```

Figure 42. EOFError Exception Handling for Invalid Duration Input

```
=== SMART LOCKER SYSTEM ===
1. View lockers
2. Store luggage
3. Retrieve luggage
4. My usage info
5. Exit

Enter option: 2

Available lockers: L02, L03, L04, L05

Enter locker ID: l02
Enter mobile number: 07123456789
Enter 4-digit passcode: 1234

Enter number of hours: xx
Handeled error: Invalid input. Please enter a number.

Enter number of hours: 1x
Handeled error: Invalid input. Please enter a number.

Enter number of hours: x1
Handeled error: Invalid input. Please enter a number.
```

Figure 43. ValueError Exception Handling for Invalid Duration Input

```
Please pay to proceed.
1. Cash
2. Card

Enter option: 1
Enter cash amount inserted: £xxx
Handeled error: Invalid amount. Please enter a number.
Enter cash amount inserted: £1x
Handeled error: Invalid amount. Please enter a number.
Enter cash amount inserted: £x1
Handeled error: Invalid amount. Please enter a number.
```

Figure 44. ValueError Exception Handling for Invalid Cash Input